

Tao Yun Huang

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Systems & Hardware Engineer

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PROFILE

Systems & hardware engineer with a solid electrical-engineering foundation and cross-domain integration experience. Hands-on across the full stack of electromechanical hardware: PCB design and hand-etching from scratch, analog assembly, and precise control of stepper motors, solenoids, and pneumatic systems. Proficient in Python, PyTorch, and C++, with embedded-systems and network protocol development, plus real-time and adversarial machine learning applied to automatic control. This span — from low-level electromechanics and embedded communication up to high-level AI algorithms and spatial sensing — lets me read the operating logic of complex systems, isolate faults precisely at the physical-signal layer under live pressure, and I aim to bring that cross-system problem-solving to precision semiconductor equipment.

PROFESSIONAL EXPERIENCE — HARDWARE ENGINEER · IF PLUS ART · 2024 — 2026

Led end-to-end development and on-site integration of complex electromechanical systems: custom PCB design and hand-etching, analog component soldering, and precise coupling of physical sensors with embedded systems. Owned motion control and hardware/software timing synchronization for a 60 kg overhead-crane armature, and routinely used instruments to isolate noise interference and communication conflicts at the physical-signal level — keeping precision equipment stable and deployable under first-line, high-pressure conditions.

Diesel-Electric Workshop

ELECTROMECHANICAL SYSTEM ·
PERMANENT INSTALL
National Railway Museum, Taipei
· 2025

60 kg armature choreographed with an overhead crane; transparent-screen overlay synced to live machinery.
Motion control · display control · multi-system timing synchronization

AI Living Lab

SENSOR HARDWARE + EMBEDDED
INTEGRATION
MODA · Taipei Computer Festival
2025
1,000+ participants · 4 days

Custom interactive-bike hardware; pedaling sensor data feeding a real-time generative-audio model I developed and trained.
Sensor hardware · embedded integration · AI model development & real-time integration

Culture-Tech Exchange

REAL-TIME NETWORKED SYSTEM
TAICCA · Pier-2, Kaohsiung
NT\$4.7B virtual investment · 4
days

Phone-based inputs synchronized to a venue leaderboard every 3 minutes across mobile ↔ venue.
Unity real-time software · cross-platform networking · live-data sync

Moss Floating Layer

MULTICHANNEL AV SYSTEM
SCOPE TECH · Shin Kong
Mitsukoshi A9 · 2025

Immersive terrarium environment in Taipei's busiest commercial district.
Multichannel spatial audio system design & integration

SELECTED ENGINEERING PROJECTS

Threshold · 閾 ELECTROMECHANICAL CONTROL SYSTEM · 3 VERSIONS v1 · Taichung 2022 v2 · C-LAB Taipei 2023 v3 · NYCU 2024	A three-version electromechanical apparatus built around a closed feedback loop. v3 drives a brass plate with stepper motors, solenoids, and pneumatics, coupled to a real-time neural-audio engine; a k-means stage redistributes output into a five-voice generative score. v2 rendered the system across a 49.4-channel Ambisonics array — the first real-time neural-audio deployment at this scale in Taiwan. <i>Solo — actuator control · feedback-loop engineering · real-time DSP pipeline · multichannel rendering</i> Stepper motors · Solenoids · Pneumatics · k-means · RAVE · Max/MSP · Ambisonics
13-Voice CV Synthesizer ANALOG CIRCUIT & PCB DESIGN 1 master · 12 sub-boards Self-fabricated · 2023	A fully self-designed, self-fabricated analog synthesizer: one master voice and twelve sub-voices, each independently CV-controlled through a shared voltage bus. PCB layouts designed from scratch and hand-etched; the architecture supports full polyphonic or multitimbral control from a single CV source. <i>Solo — circuit schematic · PCB layout & hand etching · analog assembly · system architecture</i> Analog electronics · PCB design · CV control · Hand fabrication
Latent Breach REAL-TIME VISION + UWB POSITIONING In development Submitted — ZKM Karlsruhe 2026/27	A real-time sensing system in which every smartphone becomes an instrument. An iOS / Android AR app runs on-device YOLO object detection while six UWB beacons track each device to centimeter-level precision; a PGD adversarial-perturbation pipeline is applied to detection inference as a controlled failure mode. <i>Solo — vision pipeline · UWB positioning system · adversarial ML · cross-platform AR</i> YOLO · UWB · ARKit · PGD · PyTorch · 8-ch audio
River of Shadows · 川影 IOS AR · SPATIAL RECOGNITION NYCU Administrative Building Hsinchu · 2024	A site-specific iOS AR application built on Area Target spatial recognition, aligning virtual content to a physical corridor from a first-person view and driving a synchronized spatial-audio playback layer. <i>Solo — iOS AR development · Area Target spatial recognition · spatial audio sync</i> Swift · ARKit · Area Target · Spatial audio

EDUCATION

M.F.A. Multimedia Music NYCU — National Yang Ming Chiao Tung University · Institute of Music · 國立陽明交通大學	2020 — 2024
B.Eng. Electrical Engineering Feng Chia University · 逢甲大學 · Signal processing, embedded systems, control theory	2010 — 2013
B.A. Drama & Theatre Arts · National Taiwan University of Arts · 國立臺灣藝術大學	2014 — 2019

CERTIFICATIONS

UniFi Full Stack Professional (UFSP) Ubiquiti · enterprise network design, routing & switching	2025
UniFi Wireless Admin (UWA) Ubiquiti · wireless deployment & administration	2025

TECHNICAL SKILLS

HARDWARE & CONTROL	PCB Design · Hand Etching · Analog Electronics · Stepper / Solenoid / Pneumatic Control · Motor Motion Control · Sensor Fusion
EMBEDDED & NETWORK	Embedded Systems · C++ · UDP / OSC · Network Protocols · UWB Positioning · Ubiquiti UniFi
SOFTWARE & ML	Python · PyTorch · Real-time ML · Adversarial ML (PGD) · Computer Vision (YOLO) · Unity · ARKit · Swift
SIGNAL & AUDIO	DSP · Ambisonics · Neural Audio (RAVE) · Max/MSP · Multichannel Rendering

AVAILABILITY

Immediate start · full-time

Open to relocation – Hsinchu · Taichung · Tainan · Taipei · overseas

Mandarin Chinese (native) · English (professional) · Japanese (basic)